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N + - Network Fundamentals & Building Networks.

Inter-VLAN Routing Basics.

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TASK 1.

❖ In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.

- **Creating VLANs for Different User Groups:**

Create VLANs: The following VLANs were created on the switch in Cisco Packet Tracer.

Switch(config)# vlan 10
Switch(config-vlan)# name Foodies
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Shoppers
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Streamers
Switch(config-vlan)# exit

Assign PCs to VLANs:

VLAN ID	VLAN Name	Assigned PC
10	Foodies (Zomato)	PC1 – Fa0/1
20	Shoppers (Flipkart)	PC2 – Fa0/2
30	Streamers (Spotify)	PC3 – Fa0/3

Result: Three VLANs were created, with at least one PC assigned to each VLAN.

TASK 2.

❖ Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

- **Router-on-a-Stick Configuration:**

Configure Router Sub-Interfaces

Router(config)# interface g0/0
Router(config-if)# no shutdown
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config)# interface g0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0

Assign PC IP Addresses:

PC	VLAN	IP Address	Default Gateway
PC1	Foodies (10)	192.168.10.2	192.168.10.1
PC2	Shoppers (20)	192.168.20.2	192.168.20.1
PC3	Streamers (30)	192.168.30.2	192.168.30.1

Configure Switch Port: Set the switch port connected to the router as a trunk port using switchport mode trunk.

Test Connectivity: Ping between the PCs to verify communication across VLANs.

Result: Router-on-a-stick enables PCs in all three VLANs to communicate through the router's sub-interfaces.

TASK 3.

❖ Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

- **IP Address and Default Gateway Configuration:**

The following IP addresses and default gateways were assigned to the PCs in each VLAN.

PC	VLAN	IP Address	Subnet Mask	Default Gateway
PC1	Foodies (10)	192.168.10.2	255.255.255.0	192.168.10.1
PC2	Shoppers (20)	192.168.20.2	255.255.255.0	192.168.20.1
PC3	Streamers (30)	192.168.30.2	255.255.255.0	192.168.30.1

Result: Each PC was assigned an IP address and the default gateway of its corresponding router sub-interface to enable inter-VLAN communication.

TASK 4.

❖ Change Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity, Check VLAN assignments, router sub-interface configurations, and trunk port settings.

- **Troubleshooting Inter-VLAN Connectivity:**

Identify the Problem: Users in the Shoppers (VLAN 20) could not ping users in the Streamers (VLAN 30) due to an incorrect VLAN assignment or trunk/sub-interface configuration.

Fix the Configuration:

- Checked that the PCs were assigned to the correct VLANs.
- Verified the router sub-interfaces and IP addresses.
- Ensured the switch port connected to the router was configured as a trunk.

Switch(config)# interface fa0/1
Switch(config-if)# switchport mode trunk

Verify Connectivity: Pinged from a Shoppers PC to a Streamers PC.

ping 192.168.30.2

Result: After correcting the misconfiguration, successful ping replies confirmed communication between VLAN 20 and VLAN 30.

TASK 5.

❖ Compare the router on a stick method you just configured with inter VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

- **Router on a Stick vs. Layer 3 Switch:**

Approach	Advantages (2)	Disadvantages (2)
Router-on-a-Stick	1. Cost-effective because it uses one router interface for multiple VLANs. 2. Easy to configure for small networks.	1. All VLAN traffic shares one link, which can become a bottleneck. 2. Routing performance depends on the router's capacity.
Layer 3 Switch	1. Provides faster inter-VLAN routing using hardware. 2. Handles traffic efficiently in large networks.	1. More expensive than a basic Layer 2 switch. 2. Configuration can be more complex for beginners.